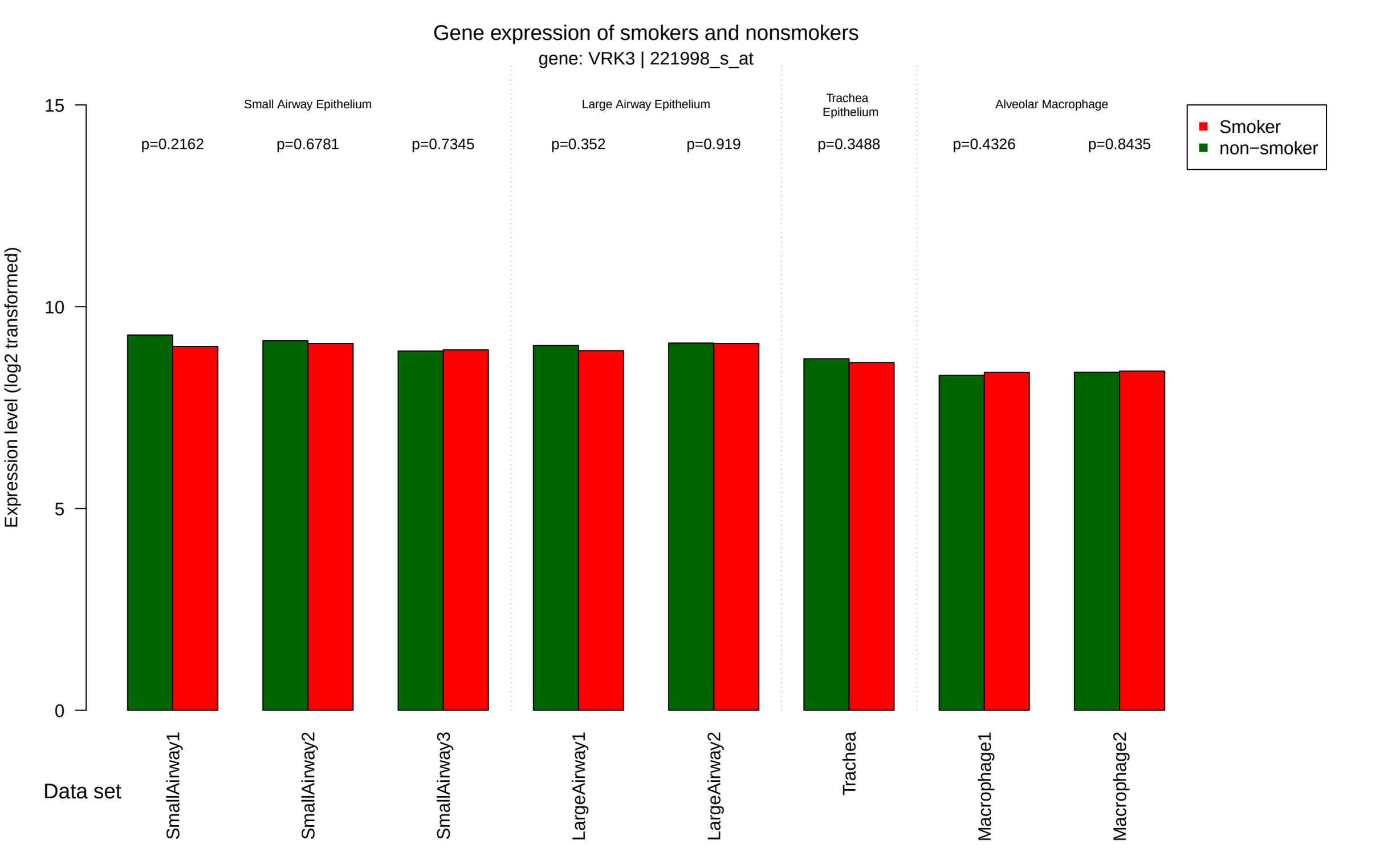
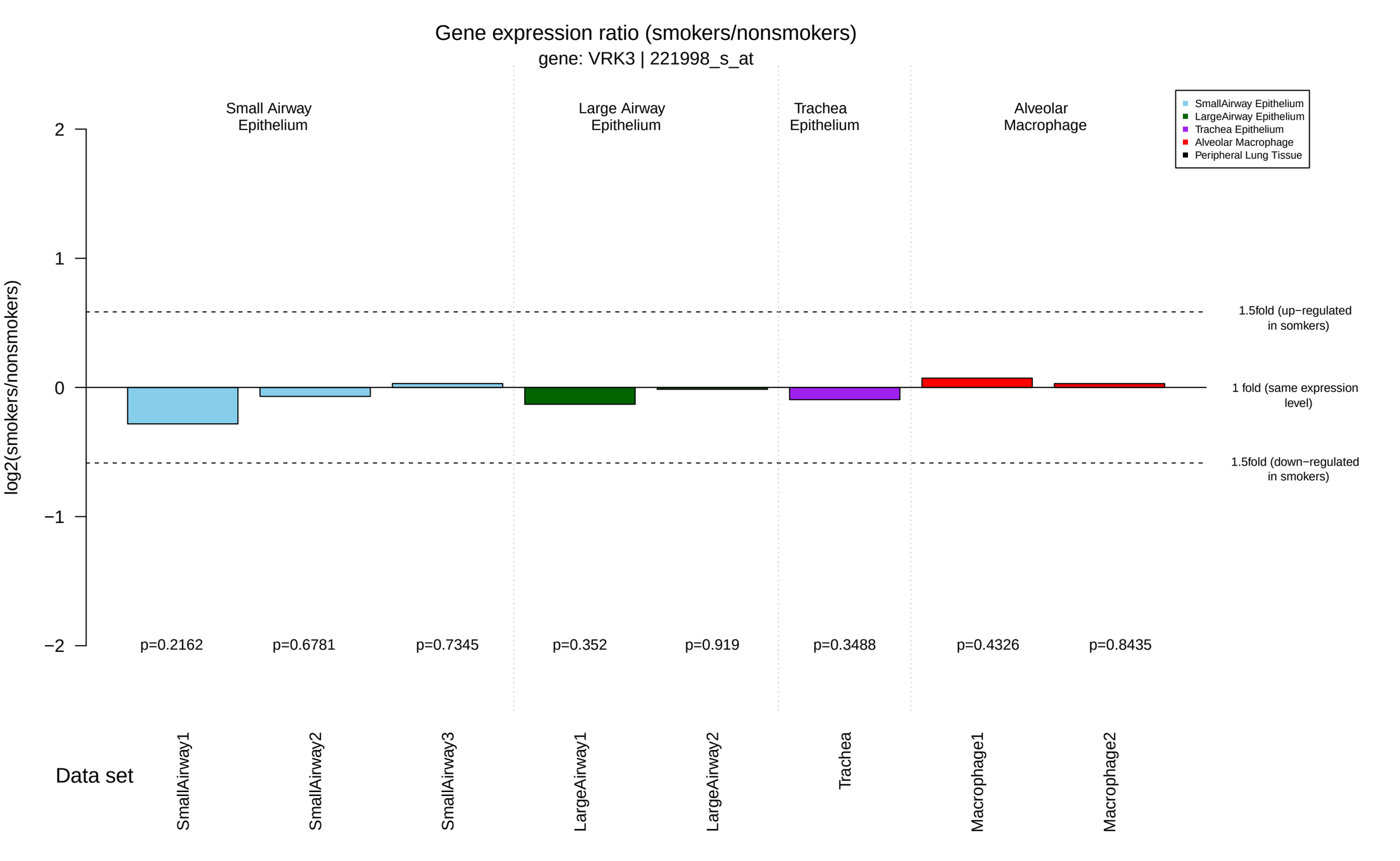
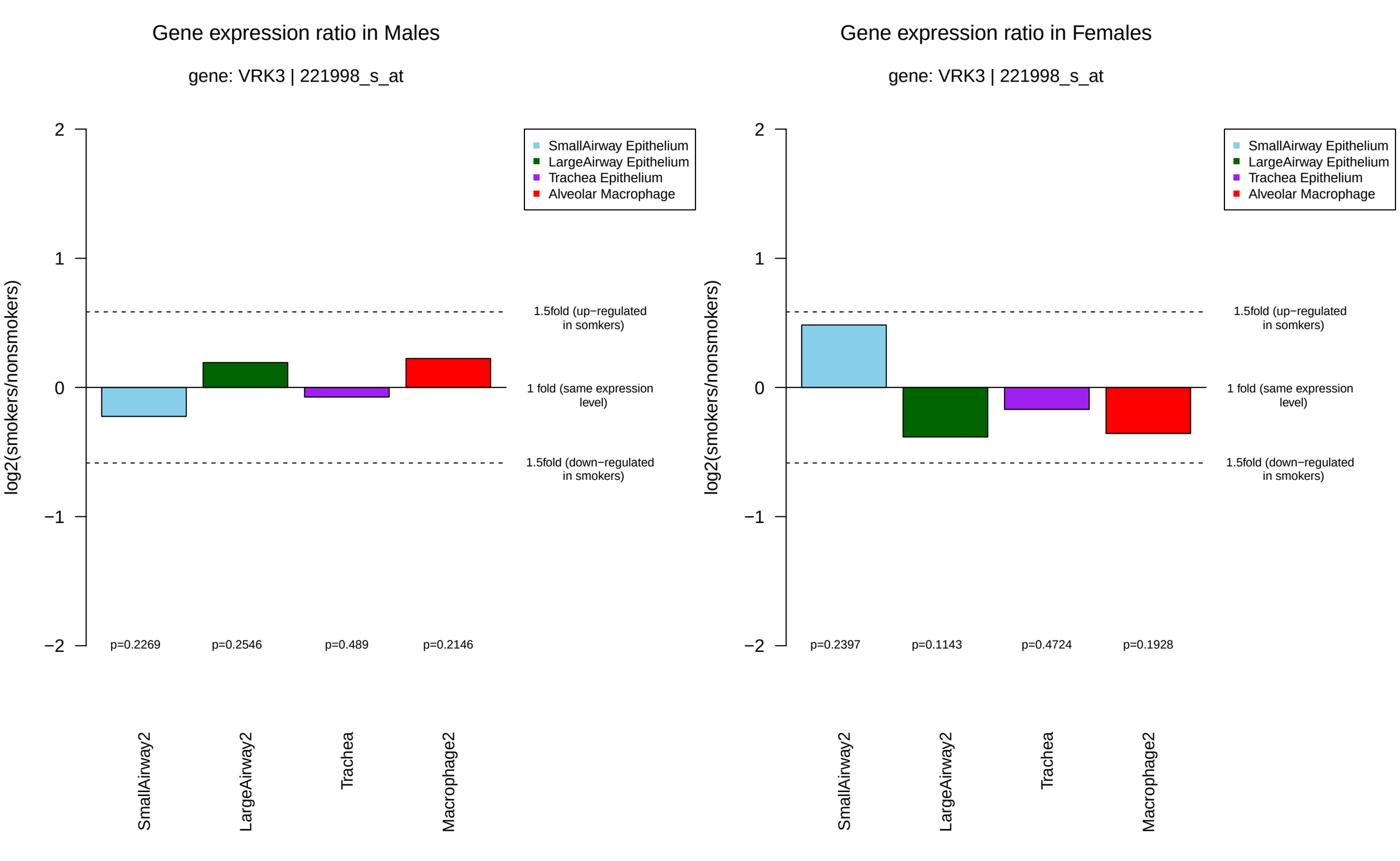
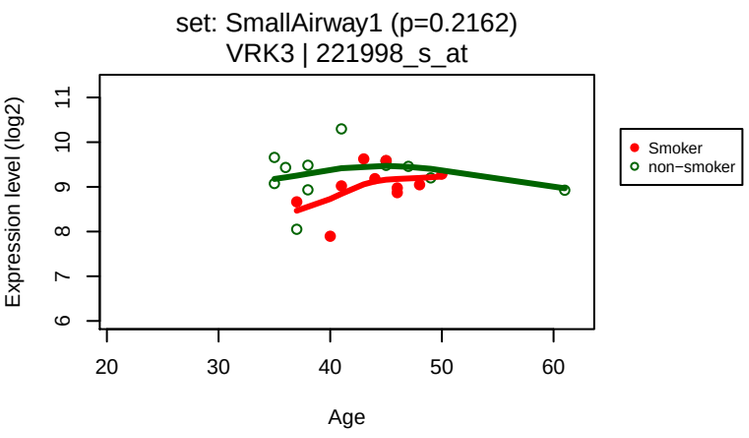
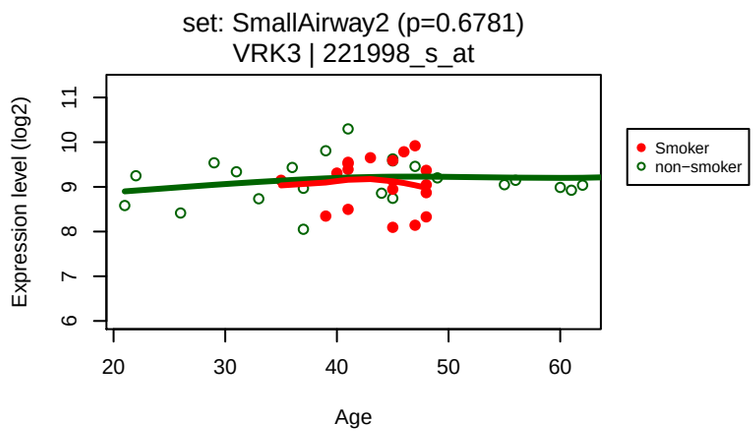
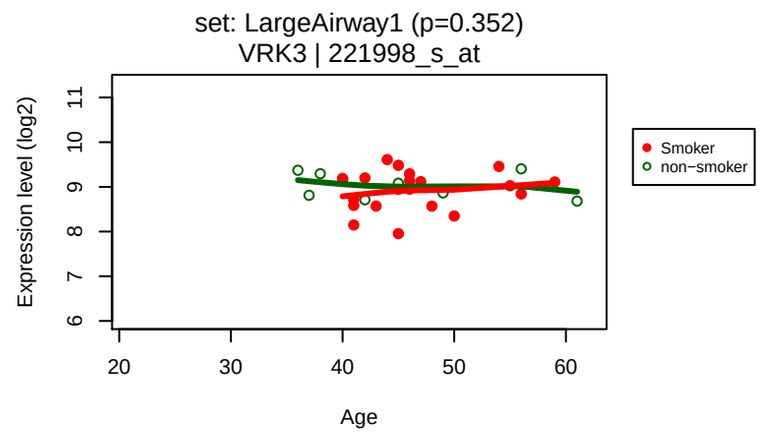

Gene expression of smokers and nonsmokers
gene: VRK3 | 221998_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2162
p=0.6781
p=0.7345
p=0.352
p=0.919
p=0.3488
p=0.4326
p=0.8435
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

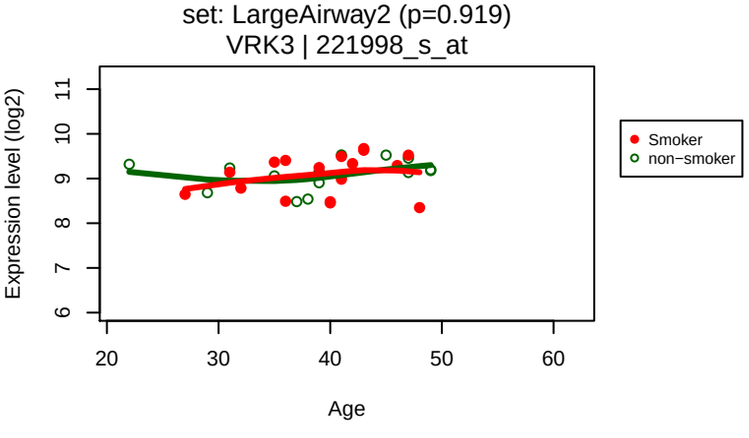

Gene expression ratio (smokers/nonsmokers)
gene: VRK3 | 221998_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2162
p=0.6781
p=0.7345
p=0.352
p=0.919
p=0.3488
p=0.4326
p=0.8435
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

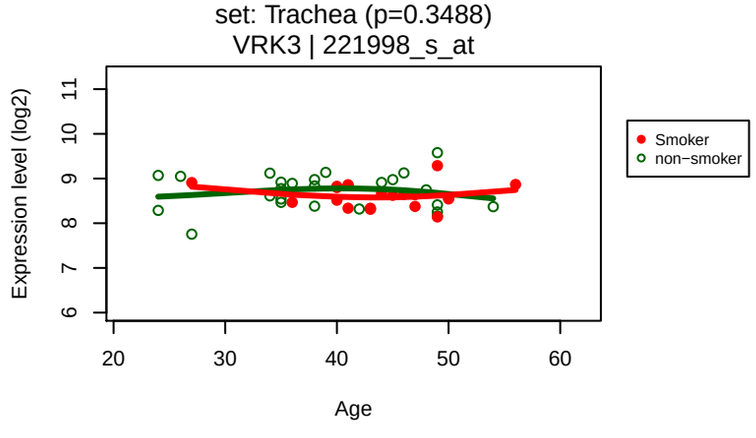

Gene expression ratio in Males
gene: VRK3 | 221998_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2269
p=0.2546
p=0.489
p=0.2146
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VRK3 | 221998_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2397
p=0.1143
p=0.4724
p=0.1928
SmallAirway2
LargeAirway2
Trachea
Macrophage2

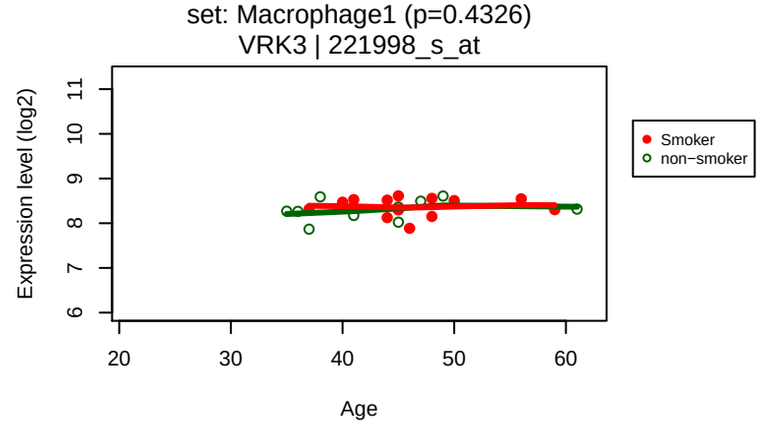

set: SmallAirway1 (p=0.2162)
VRK3 | 221998_s_at
Expression level (log2)
Age
Smoker
non−smoker

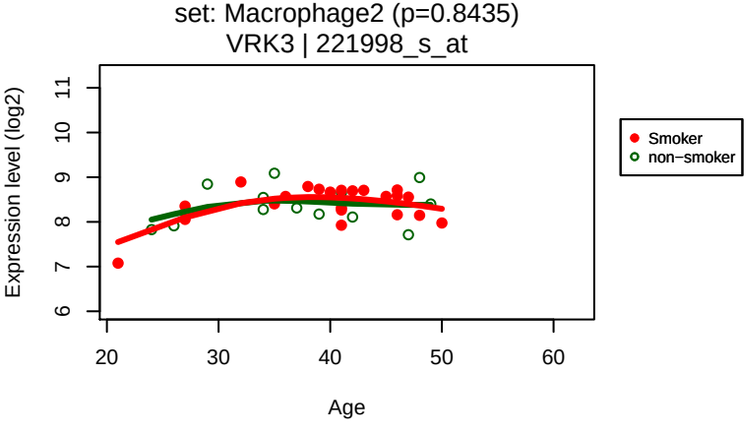

set: SmallAirway2 (p=0.6781)
VRK3 | 221998_s_at
Expression level (log2)
Age
Smoker
non−smoker

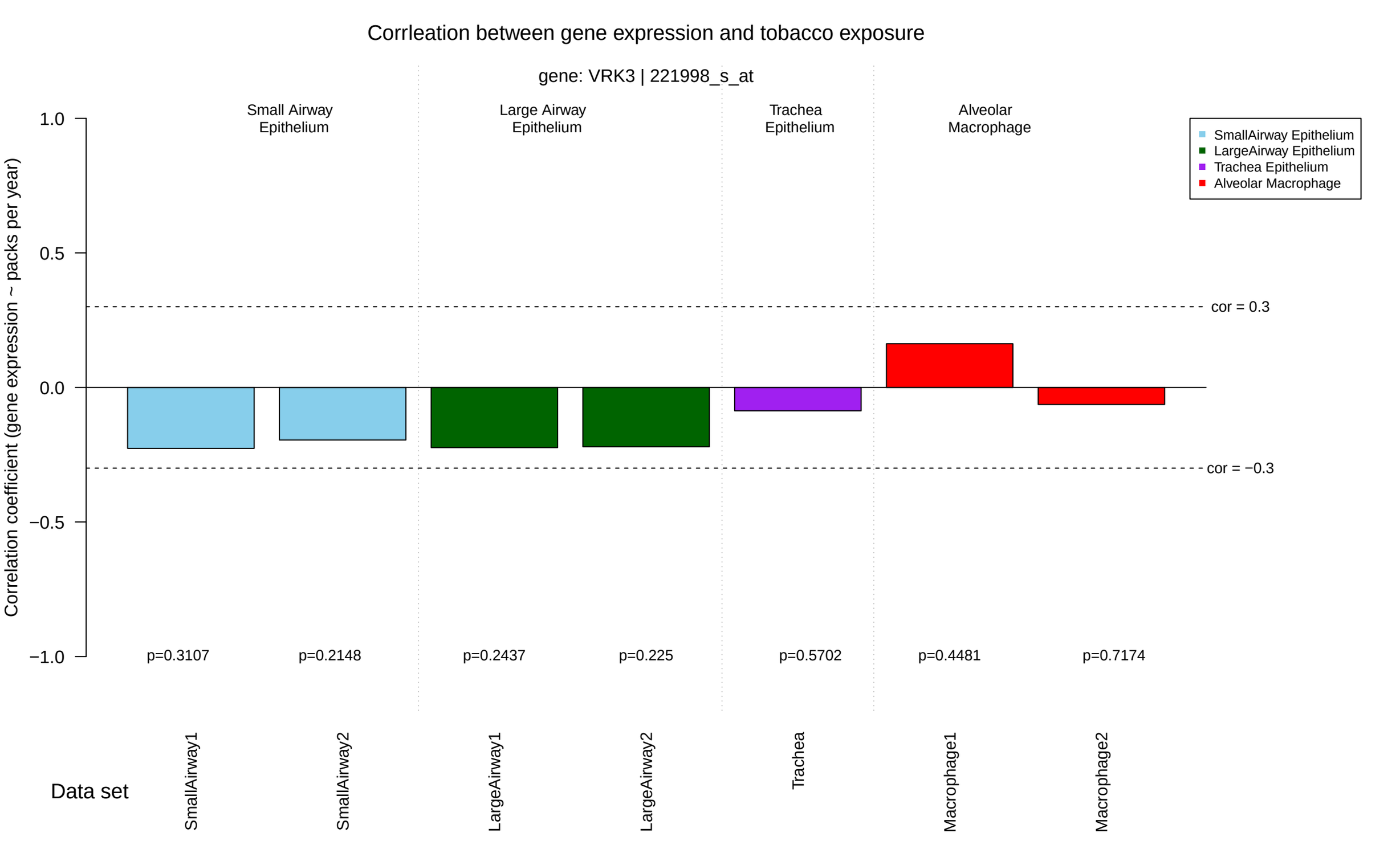

set: LargeAirway1 (p=0.352)
VRK3 | 221998_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.919)
VRK3 | 221998_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3488)
VRK3 | 221998_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4326)
VRK3 | 221998_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8435)
VRK3 | 221998_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VRK3 | 221998_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3107
p=0.2148
p=0.2437
p=0.225
p=0.5702
p=0.4481
p=0.7174
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2